BUSINESS INTELLIGENCE LAB

EXPERIMENT 8

Report Scheduling, Automation and Alerts using Microsoft Power BI

Report

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1. AIM

To design and implement report scheduling, automation, and alert mechanisms using Microsoft Power BI and develop strategies for automated report delivery, monitoring, and business performance tracking.

2. INTRODUCTION / THEORY

Business intelligence reporting helps organizations monitor performance and make informed decisions. Scheduled reports provide information to users at planned intervals, while distribution matrices identify the appropriate audience and delivery method. KPI alerts help users identify unusual or important changes in business performance. Power BI Desktop is used to create reports, while Power BI Service supports publishing, dashboards, and eligible data alerts.

3. SOFTWARE / TOOLS REQUIRED

- Microsoft Power BI Desktop
- Microsoft Power BI Service
- Global Superstore Dataset.xlsx
- Microsoft Excel

4. IMPLEMENTATION AND OBSERVATIONS

4.1 Scheduled Reporting Plan

The following reporting plan was prepared to define the frequency and intended users.

Report	Frequency	Intended Users
Daily Sales Report	Daily	Sales Team and Sales Manager
Weekly Profit Report	Weekly	Regional Managers and Finance Team
Monthly Executive Dashboard	Monthly	Senior Management

4.2 Report Distribution Matrix

User Group	Report	Frequency	Delivery Method
Sales Team	Daily Sales Report	Daily	Email
Regional Managers	Weekly Profit Report	Weekly	Email / PDF
Finance Team	Weekly Profit Report	Weekly	Shared Workspace
Senior Management	Monthly Executive Dashboard	Monthly	Power BI Service

4.3 Publishing to Power BI Service

The report was published successfully from Power BI Desktop. The report and its semantic model were visible in My workspace.

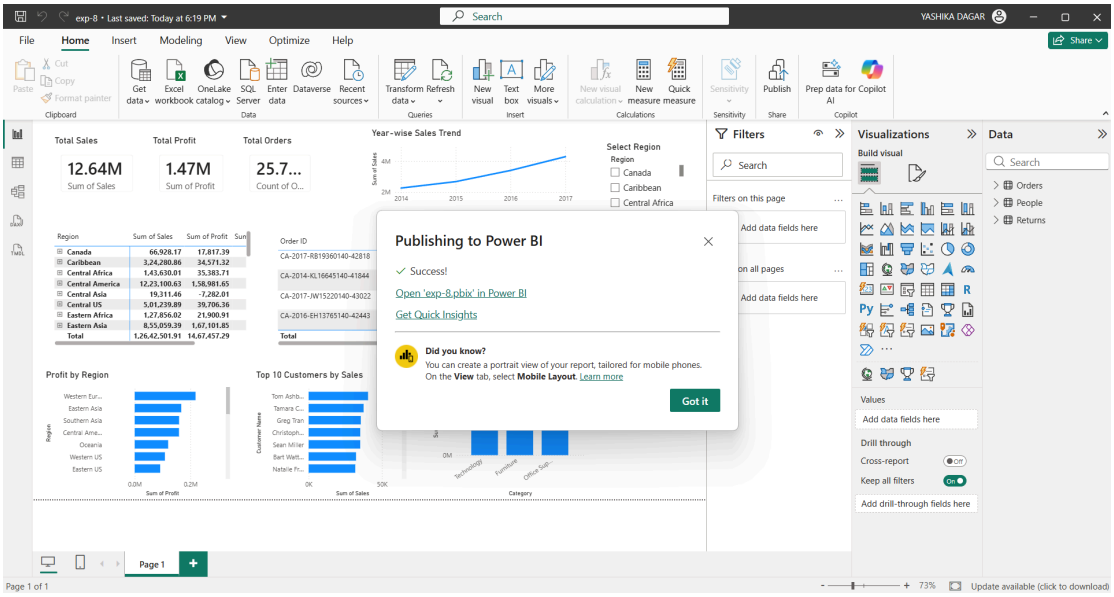


Figure 1: Successful publishing confirmation.

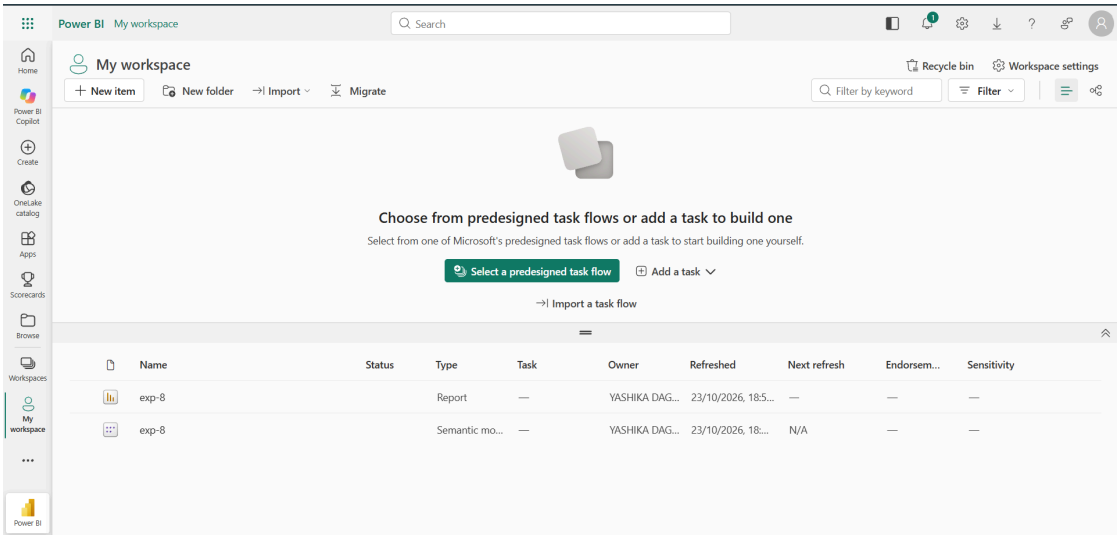


Figure 2: Report and semantic model available in My workspace.

4.4 KPI Dashboard and Alert Configuration

A dashboard named Experiment 8 KPI Monitoring was created. KPI visuals were pinned to the dashboard for monitoring.

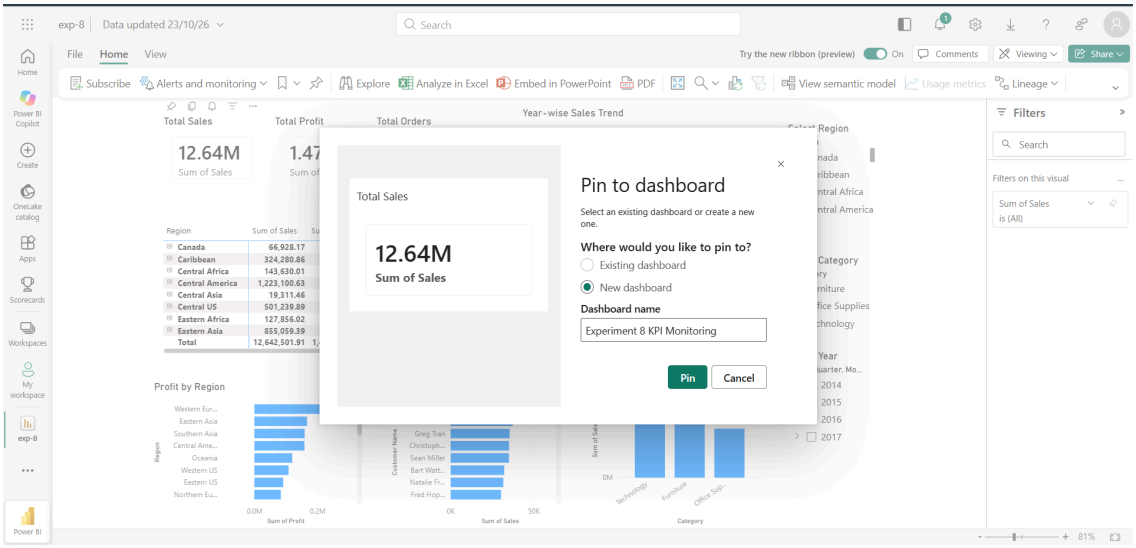


Figure 3: Pinning the Total Sales KPI to a new dashboard.

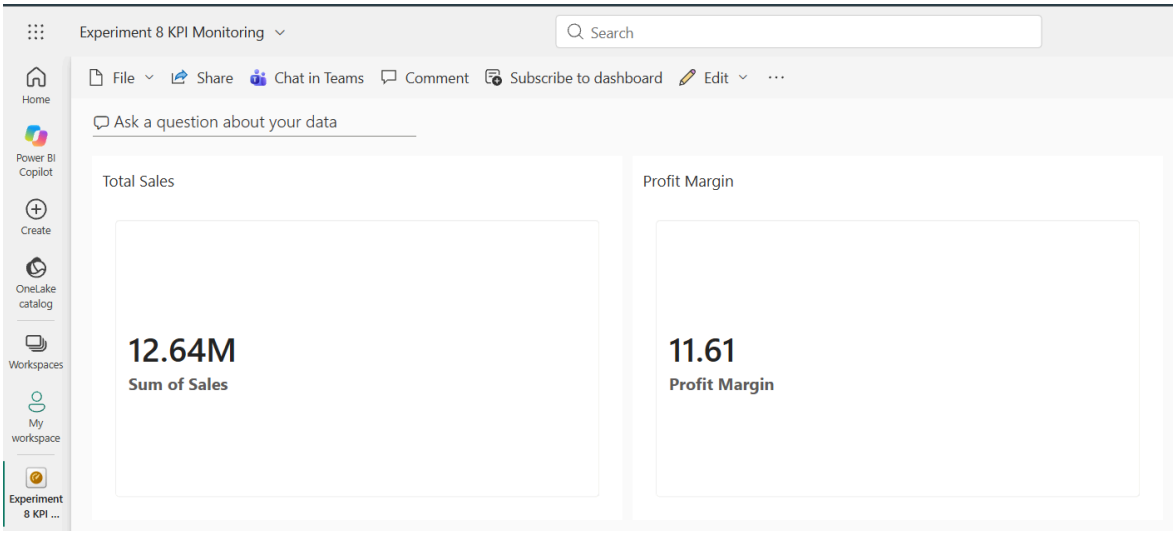


Figure 4: Dashboard containing Total Sales and Profit Margin KPI tiles.

The following fixed-threshold alert rules were configured:

Alert	Configured condition	Interpretation
Sales Performance Alert	Sum of Sales Above 12,642,501.91	Fixed-value threshold; not a direct 15% decline calculation.
Profit Margin Alert	Profit Margin Below 10	Matches the specified below-10 KPI threshold.
Returns Monitoring Alert	Total Returns Above 2,400	Fixed-value threshold; not a dynamic 20% increase calculation.

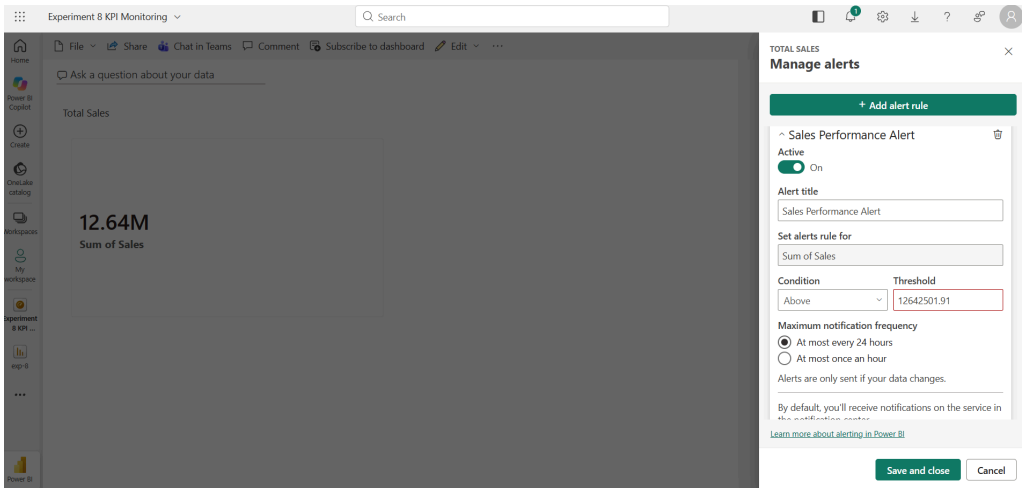


Figure 5: Sales Performance Alert settings.

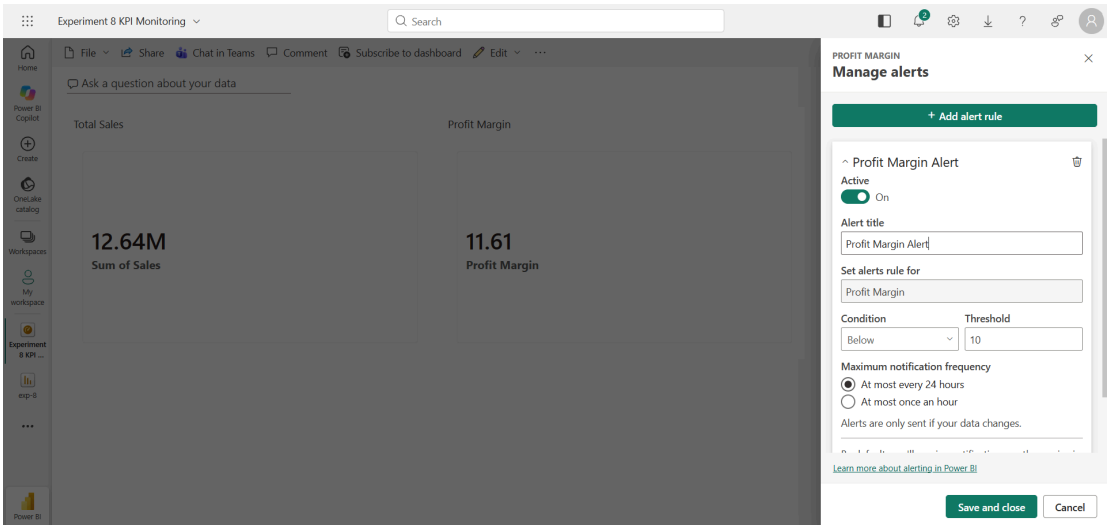


Figure 6: Profit Margin Alert settings.



Figure 7: Total Returns KPI card.

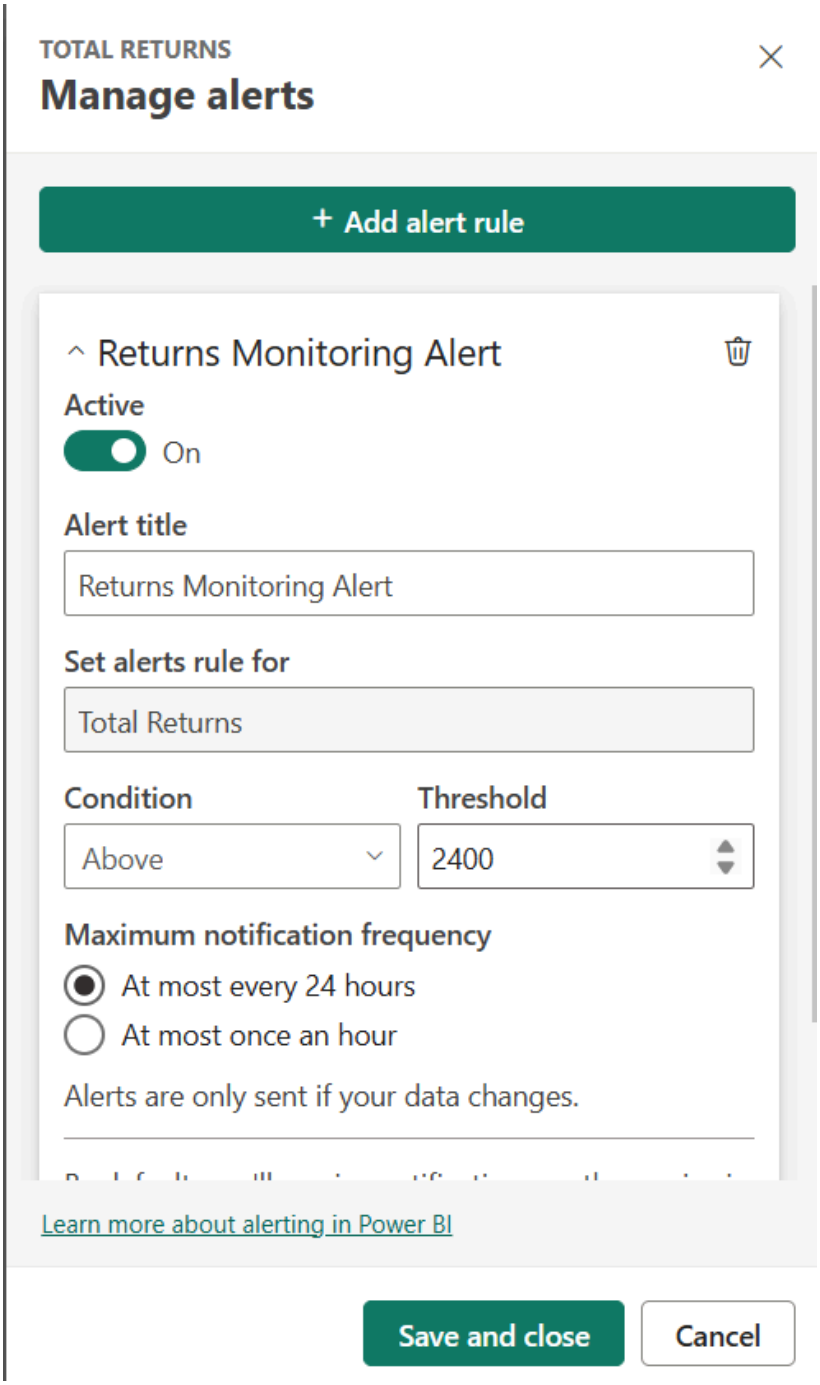


Figure 8: Returns Monitoring Alert settings.

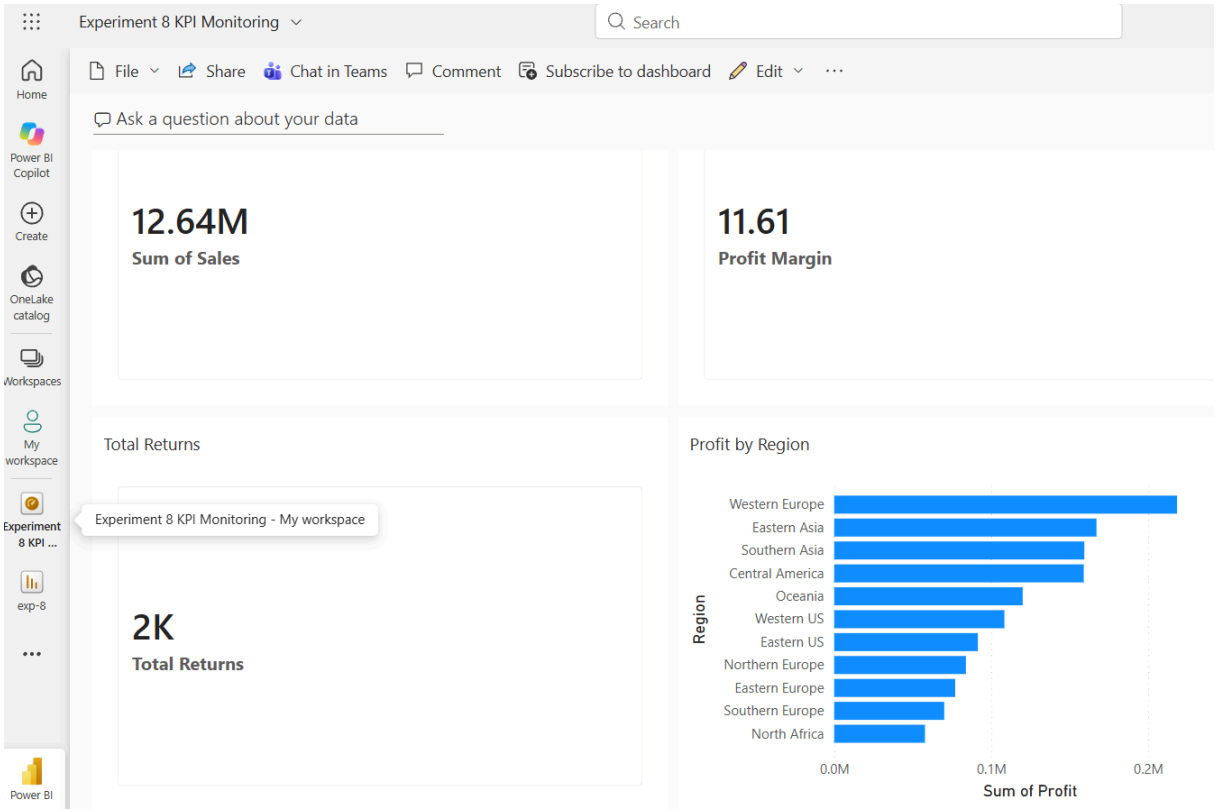


Figure 9: Final KPI Monitoring dashboard including Profit by Region.

4.5 Monitoring Scenario and Recommendations

Scenario: A regional branch experiences declining profitability. The report's Profit by Region visual and regional sales/profit data can be used to identify weaker-performing regions and investigate the causes.

- Monitor Profit Margin and investigate when it falls below 10%.
- Review sales trends and investigate a decline greater than 15% against the previous comparable period.
- Review return counts and investigate an increase greater than 20% against the previous comparable period.
- Assign the issue to the Finance Manager, Sales Manager, Operations Manager, and relevant Regional Manager.
- Review discount strategies, low-performing products, category profitability, operating costs, and sales recovery actions.

Power BI's configured alerts shown here use fixed numeric thresholds. The percentage-change conditions in the experiment brief are retained as monitoring rules for comparison with a previous period; the fixed alerts should not be interpreted as automatically calculating those percentage changes.

4.6 Scheduling / Subscription Limitation

When attempting to access a paid Power BI feature related to sharing/subscription, the service displayed an “Upgrade to a paid Power BI license” message. Therefore, the scheduled reporting plan and

distribution matrix were designed and documented, but recurring email delivery/subscription was not activated in this account.

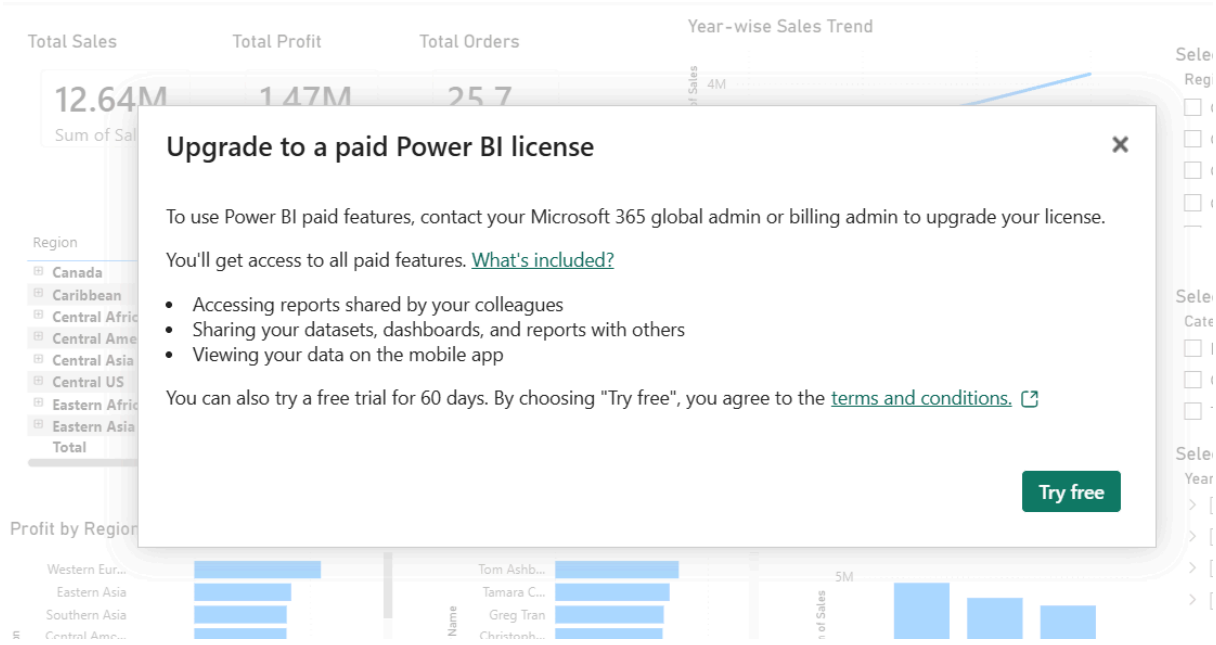


Figure 10: Power BI Service license-upgrade prompt.

5. RESULT

The Power BI report was successfully published to Power BI Service and an Experiment 8 KPI Monitoring dashboard was created. Total Sales, Profit Margin, Total Returns, and Profit by Region visuals were included. Fixed-threshold alerts were configured for the three KPI tiles, while the scheduled reporting plan, distribution matrix, and regional monitoring recommendations were documented.

6. CONCLUSION

This experiment demonstrated how Power BI Desktop and Power BI Service can be used together for report publishing, KPI dashboard creation, and alert-based monitoring. It also highlighted the importance of distinguishing fixed-value alerts from percentage-change rules and the role of licensing in enabling certain automated delivery features.

7. LEARNING OUTCOMES

- Designed a reporting schedule for daily, weekly, and monthly reports.
- Prepared a report distribution matrix for different business user groups.
- Published a report to Power BI Service and created a monitoring dashboard.
- Configured fixed-threshold alerts on supported dashboard tiles.
- Analyzed a regional profitability monitoring scenario and proposed corrective actions.

8. REFERENCES

- Microsoft Power BI Desktop
- Microsoft Power BI Service
- Global Superstore Dataset.xlsx
- Experiment 8 laboratory instructions